

Voice Emergency Checklist System

Generic advisory reader · audio-panel input · Config + Data cards

One combined reference: product, Aircraft Card spec, certification basis, hardware, and enclosure. Supersedes the separate tech, processor, and enclosure documents.

Cert path: NORSEE (PS-AIR-21.8-1602) · DO-160G environmental · deliberate DO-178C-avoidance argument · STC/PMA placement.

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS

Not airworthy and not certified. The certification sections describe the path a productized unit would follow — a plan, not held approvals. No NORSEE LOA, STC, PMA, or TSOA exists for this article.

Scope & Status

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS. This document describes a bench / training prototype and a certification-readiness roadmap. As built, the unit is not airworthy, not certified, and must never be installed in or relied upon aboard an aircraft. Nothing here is an STC, TSO authorization, PMA, or NORSEE letter of approval. The certification sections (Part C) describe the path a productized unit would follow — they are a plan, not a claim of compliance.

The product is a generic, offline, voice-driven advisory emergency-checklist reader. The pilot states an emergency by name; the device reads each checklist item aloud and waits for a spoken completion word before advancing. It runs entirely on-device (no cloud, no Wi-Fi). The product itself is aircraft-agnostic — all aircraft-specific behavior comes from the installed cards (Part B). The Cessna Citation CJ2 is included only as the reference example.

Crew audio is taken from the aircraft audio panel (analog or digital, selectable per installation), and recognition is voice-activated (VOX) by default — the pilot simply speaks, with a push-to-talk (PTT) button retained as a manual override. Configuration and checklist data live on two separate cards — a write-protected Config Card that defines the installation (aircraft selection, audio source, VOX behavior, hardware options) and a Data Card that carries the checklist library and audio. This two-card, audio-panel-fed design is a deliberate change from the earlier onboard-microphone / single-card demo and carries certification consequences addressed honestly in Part C.

This master reference supersedes and combines the previously separate documents: the technical data packet, the processor-selection note, and the enclosure specification. Those remain in the repository for history; this is the single source of truth.

Part	Contents
A Product	What the system is, the generic architecture, and the safety model
B The cards	The two-card (Config + Data) model + the formal card specifications & validation
C Certification basis	NORSEE / DO-160G / installation path, the deliberate DO-178C-avoidance argument, and the audio-panel-interface impact
D Hardware reference	Pin map, audio-input stage, VOX/PTT, per-device wiring, annunciator, lamp driver, power, BOM, processor selection
E Enclosure	Two-piece mechanical specification (dual card slots, isolated audio interface) for the fabricating engineer
F References	All cited regulatory and component sources

PART A

The Product

A generic, offline, advisory checklist reader. This part defines what the system is (and is not), its block-level architecture, and the three safety rules that the entire certification argument in Part C is built on.

A.1 What it is (and is not)

It is	It is not
An advisory read-aloud reader of checklist items	A required or primary aircraft system
Receive-only on a single audio tap; no command/data to any aircraft system	A transmitter, a panel control, or an interface to avionics/engines/controls
Driven entirely by the installed Config Card + Data Card	Tied to one airframe in firmware
Offline, deterministic, single-chip	A cloud / connected / large-vocabulary STT device
A complement to the certified/required checklist	A substitute for the AFM/QRH or required checklist

This framing is not cosmetic — it is the foundation of the certification argument in Part C. A device that is non-required, advisory-only, fails to a clearly-annunciated safe state, and connects to the aircraft only through a galvanically-isolated, receive-only audio tap is the profile that fits the NORSEE (Non-Required Safety Enhancing Equipment) approval path and that lets the program lean on DO-160G environmental qualification while avoiding DO-178C software assurance. Note that the audio-panel tap is a wired interface to an aircraft system — it deliberately trades the old “electrically independent” claim for a weaker but defensible “receive-only, isolated” posture. Part C addresses this honestly; it is the most significant certification change in this revision.

A.2 Generic system architecture

Block	Function	Install-/aircraft-specific?
MCU + speech stack	VOX/wake word → command recognition → playback sequencing	No — fixed firmware
Audio-panel input stage	Takes crew speech from the aircraft audio panel — analog (isolated line tap) or digital (I2S codec), selectable per install	Config-driven — source set by the Config Card; path wired at install
Speaker + amplifier (I2S)	Reads checklist items / annunciations aloud (own speaker; not fed back to the panel)	No
Config Card (microSD, slot 1)	Defines the installation: active aircraft, audio source (analog/digital), VOX parameters, hardware options	Yes — per installation
Data Card (microSD, slot 2)	Carries the checklist library, trigger/advance vocabulary, and read-aloud audio	Yes — per aircraft
Annunciator switch	Dark-cockpit status / fault indication, IN/OUT select, PTT override	No

The MCU runs Espressif ESP-SR (AFE noise-suppression/VAD → WakeNet wake word → MultiNet fixed-grammar command recognition). The AFE's voice-activity detector (VAD) is what enables hands-free VOX: the pilot speaks and the device gates recognition on detected speech, with the PTT button retained as a manual override (force-listen). The grammar (trigger phrases, advance words) is small and bounded, which is what keeps an MCU-class part in scope (see D.7).

The crew-audio source is no longer an onboard microphone in the operational design — it is a tap off the aircraft audio panel. An onboard MEMS microphone is retained only as a documented bench-test option (D.3), never as the installed audio source.

A.3 The safety model (carried into the cert argument)

Three design rules define the failure behavior, and each one maps to a NORSEE requirement (Part C):

- 1 Revert-to-unopened. If the card is missing, unreadable, malformed, fails schema validation, or any referenced audio clip is absent, the device refuses to present any checklist and enters a clearly-annunciated FAULT state. It never shows partial or stale data. (*Failure mode = loss of function, not misleading information.*)
- 2 Dark-cockpit annunciation. When selected IN and healthy, the unit shows nothing. A fault lights the amber FAULT legend. Selected OUT shows the white OFF status and inhibits the fault legend (a deliberately deselected system needs no crew action). A power-up lamp test proves the legend is alive.
- 3 Receive-only, isolated interface. The device's only tie to an aircraft system is the audio-panel input, and that tie is one-way (listen) and galvanically isolated (see C.3a, D.3.1): a high-impedance, isolation-transformer-coupled tap for the analog path, or a buffered receive-only feed for the digital path. The unit cannot transmit, key, mute, or back-feed the panel, and it still draws power on its own protected rail. A short, open, or failure of the device cannot affect audio-panel function. This replaces the former “no electrical tie at all” claim with a narrower, testable one — and it is the crux of the Part C argument.

PART B

The Two Cards

Card defines everything — split across two. The product is generic; two installed microSD cards make it a specific aircraft's checklist reader, installed a specific way. A write-protected Config Card describes the installation; a Data Card carries the checklist content. This part is the formal specification — layouts, schemas, voice-grammar rules, and the validation logic that implements revert-to-unopened.

B.1 Principle

The product is generic. The two installed cards make it a specific aircraft's checklist reader, installed a specific way. Responsibility is split: the Config Card describes this installation (which aircraft, which audio source, how VOX behaves, what hardware is fitted); the Data Card carries the checklist content (library, vocabulary, audio). To support a new airframe you author a new Data Card; to re-use it in a different aircraft or wiring you change only the Config Card. No firmware change, no recompile.

Why two cards? Configuration is installation-controlled (set by the installer/shop and locked) while checklist data is content-controlled (authored from the AFM/QRH and revised as the source revises). Separating them keeps a content revision from silently changing the installation's audio/VOX setup, and lets a write-protected Config Card serve as the configuration-control record the FAA expects under NORSEE (Part C). Both cards are microSD/FAT32 and sit in two physical slots (slot 1 = CONFIG, slot 2 = DATA; see E.3).

B.2 Config Card (slot 1) — layout & schema

```
/sdcard-config/ (slot 1, microSD, FAT32)
config.json      the single installation-configuration manifest
```

config.json top-level object:

Field	Type	Req.	Meaning
schema_version	integer	rec.	Config-schema version the firmware validates against
aircraft	string	yes	Active aircraft id; must match a folder on the Data Card
audio_source	string	yes	analog or digital — selects the audio-panel input path (D.3.1)
audio	object	yes	Audio-input parameters (below)
vox	object	yes	VOX behavior (below)
hardware	object	opt.	Fitted-hardware options (codec part, legend rail, PTT present, etc.)
install	object	rec.	Provenance: shop, installer, date, work-order (configuration control)

audio object:

Field	Type	Req.	Meaning
input_gain_db	number	rec.	Input trim for the line-level tap (typ. 0–6 dB)

Field	Type	Req.	Meaning
codec	string	when digital	I2S codec fitted (e.g. es8388, es7210, pcm1808)
sample_rate_hz	integer	rec.	16000 for ESP-SR

vox object:

Field	Type	Req.	Meaning
mode	string	yes	vox (default, hands-free) or ptt_only (override-only)
vad_sensitivity	integer	rec.	AFE VAD aggressiveness 0–3 (higher = less false-trigger, may clip onset)
hangover_ms	integer	rec.	How long to keep listening after speech stops (debounce, typ. 300–600 ms)
ptt_override	boolean	rec.	true keeps PTT as a force-listen override even in vox mode

- ptt_only mode disables VOX and reverts to the legacy push-to-talk behavior.
- A Config Card whose aircraft has no matching Data Card folder is a fault (FAULT_AIRCRAFT_MISMATCH), not a silent guess.

B.3 Data Card (slot 2) — layout

```
/sdcard-data/  (slot 2, microSD, FAT32)
/              one folder per aircraft (e.g. CJ2, B200, PC12)
  checklists.json  the data manifest + checklist library
  audio/
    .wav           one read-aloud clip per referenced item + fault clips
```

- The firmware loads the folder named by the Config Card's aircraft field. If no Config Card is present at all, that is FAULT_NO_CONFIG — the device does not fall back to guessing a folder.

B.3a Data Card manifest schema (checklists.json)

Top-level object:

Field	Type	Req.	Meaning
schema_version	integer	rec.	Card schema version the firmware validates against
aircraft	string	yes	Short aircraft id; must match the folder name
title	string	yes	Human-readable aircraft name shown in logs
universal_advance	string[]	yes	Advance words accepted on every item (e.g. check, checked, complete, next)
checklists	object[]	yes	The checklist library (≥ 1)

Each checklists[] entry:

Field	Type	Req.	Meaning
id	string	yes	Stable identifier (e.g. engine_fire)
title	string	yes	Spoken/displayed checklist name
type	string	rec.	emergency, abnormal, normal
triggers	string[]	yes	Phrases that invoke this checklist by voice
items	object[]	yes	Ordered checklist steps (≥ 1)

Each items[] entry:

Field	Type	Req.	Meaning
clip	string	yes	Audio basename in audio/ (read-aloud <clip>.wav)
text	string	yes	The item text (for logs / optional display)
advance	string[]	opt.	Item-specific completion words (in addition to universal_advance)

B.3.1 Voice-grammar rules (validation-enforced)

MultiNet (English) imposes grammar constraints that the card author must honor; the validator rejects a card that violates them (revert-to-unopened):

- Phrases are lowercase letters and single spaces only — no digits or punctuation.
- Spell numbers as words: “V-one” → v one, not V1.
- Keep the total command count modest (well under the ~200-command MultiNet cap).
- Every clip referenced by any item must exist as audio/<clip>.wav.

B.4 Two-card validation & the revert-to-unopened rule

On boot the firmware attempts a complete, valid load of both cards and returns exactly one status. Both cards must be present, valid, and mutually consistent (the Config Card's aircraft must resolve to a Data Card folder):

Status	Card	Meaning	Annunc.
STORE_OK	both	Both cards mounted, both manifests parsed + schema-valid, every audio clip present, aircraft consistent	dark
FAULT_NO_CONFIG	config	Config Card not detected / config.json missing / unreadable	amber
FAULT_CONFIG_PARSE	config	config.json malformed	amber
FAULT_CONFIG_VALIDATION	config	Config schema / value violation (bad audio_source, bad vox.mode, missing required field)	amber
FAULT_NO_CARD	data	Data Card not detected / mount failed	amber
FAULT_AIRCRAFT_MISMATCH	both	Config aircraft has no matching Data Card folder	amber
FAULT_NO_JSON	data	checklists.json missing / unreadable	amber
FAULT_PARSE	data	Data JSON malformed	amber
FAULT_VALIDATION	data	Data schema or voice-rule violation, empty data	amber
FAULT_AUDIO_MISSING	data	A referenced clip is absent	amber
FAULT_NO_MEMORY	—	Allocation failed while loading	amber

On any FAULT the in-memory checklist table is left EMPTY — the device cannot present a partial or stale checklist even if asked. This is the literal implementation of “always revert to unopened if files are not available.” The boot order is Config Card first (it names the aircraft and the audio source the input stage must initialize), then the matching Data Card folder.

B.5 Card integrity & provenance (recommended for productized cards)

For cards that drive a *safety-enhancing* device, integrity matters as much as schema validity.

Recommended additions (forward-looking, beyond the current demo):

- Schema version gate — firmware refuses either card whose schema_version it does not support, rather than mis-parsing it.
- Manifest checksum / signature — a per-card hash (and, for production, a signature) so a corrupted or tampered card is rejected (FAULT_CONFIG_VALIDATION / FAULT_VALIDATION).
- Content provenance (Data Card) — record the AFM/QRH source revision the checklist text was transcribed from, the author, and the date.
- Installation provenance (Config Card) — the install object records shop, installer, date, and work-order. The Config Card is the per-tail configuration-control record the FAA expects under NORSEE (Part C): it documents *how this specific aircraft was set up*, including the audio source and VOX behavior.
- Read-only media — distribute production cards write-protected; the Config Card in particular should be locked after installation so configuration cannot drift in service.

PART C

Certification Basis

STC / PMA / NORSEE-ready reference. This part is a roadmap, not evidence of compliance — the demo unit holds no approvals. It explains why NORSEE is the right path, the standards elected as Minimum Design Requirements, and the deliberate, documented argument for avoiding DO-178C.

Status: this part is a roadmap, not evidence of compliance. The demo unit holds no approvals. The intent is to design and document so that a productized unit can pursue approval efficiently.

C.1 Why NORSEE is the right path

The product profile — non-required, advisory, independent of primary systems, failing to a clearly-annunciated safe state — matches the FAA's Non-Required Safety Enhancing Equipment (NORSEE) policy, PS-AIR-21.8-1602 (issued 03/31/2016)¹. NORSEE approval is a combined design and production approval issued under 14 CFR § 21.8(d) for equipment that is “not required by any Federal regulation with the intent to measurably increase aircraft safety,” determined to be a minor change to type design with a minor failure condition. A voice reader that simply reads aloud the same items already in the certified/required checklist — while the paper/electronic required checklist remains the authority — is a strong fit.

C.1.1 Applicability limit (read this)

NORSEE policy applies to aircraft certified under 14 CFR Part 23, 27, and 29 (and predecessor categories) and explicitly excludes Part 25 transport-category aircraft¹.

- The reference example, the Citation CJ2 (525A), is a Part 23 airplane — so NORSEE is available for it.
- Because the product is generic, the card library will inevitably include Part 25 aircraft (e.g. larger jets). For a Part 25 installation, NORSEE is not available; that installation must use an STC (or other major/minor-change path appropriate to the airframe). The product's certification status is therefore per-airframe, and the documentation must say so explicitly.

C.2 Minimum Design Requirements (the standards we elect)

Under NORSEE the applicant proposes the industry standard(s) that become the Minimum Design Requirements (MDR); the FAA accepts, partially accepts, or augments them. The FAA recommends widely-accepted RTCA / SAE / ASTM standards¹.

Topic	Elected standard / basis	Notes
Environmental	RTCA/DO-160G	Primary qualification evidence. Categories in C.4. FAA accepts DO-160 D/E/F/G per AC 21-16G and “strongly encourages DO-160G for new articles.”
Software	No DO-178C sought — see C.3	Argued out via function/failure classification, not avoided by omission
Complex hardware	DO-254 not invoked	COTS MCU + simple discrete logic; no custom complex devices (ASIC/FPGA/PLD). Per AC 20-152A, <i>simple</i> hardware verifiable by test needs no DO-254 design assurance.

Topic	Elected standard / basis	Notes
Human factors / color	FAA AC 25-11B conventions	amber=caution, white=status, dark-cockpit. Accepted design convention even though the unit is not a Part 25 article.
System safety (if needed)	SAE ARP4761 / AC 23.1309-1	Only invoked if the safety evaluation ever rises above "minor" (it should not)

C.3 The DO-178C–avoidance argument (deliberate, documented)

DO-178C is software *design-assurance* guidance whose rigor scales with the failure condition the software can contribute to. NORSEE lets us classify the function and, for a genuinely minor–failure advisory device, scope software assurance down to essentially nothing formal — provided the architecture earns it. The argument has four legs:

- 1 Function is advisory, not required. The device does not perform, command, or feed any required function. The crew's authority is the certified/required checklist; this unit only reads it aloud.
- 2 Worst-case failure is minor. The two credible failures are (a) loss of function — it goes silent or annunciates FAULT, and the crew uses the paper/QRH exactly as today; and (b) misleading information — mitigated by the revert-to-unopened rule (no partial/stale output), card validation, and a procedural requirement that the crew cross-checks against the required checklist. Neither failure reduces the crew's ability to cope with a condition worse than minor — the NORSEE safety-evaluation test¹.
- 3 Bounded interface (not full independence). This revision adds one tie to an aircraft system: a receive-only, galvanically-isolated audio tap off the audio panel (C.3a). The device takes no input that commands a function and produces no output to any aircraft system — it only listens. Physical/electrical separation is preserved on the power and signal–return side by the isolation barrier. This is a weaker claim than the former “no electrical tie at all,” and is called out as such; the argument now rests on *directionality + isolation* rather than total separation¹.
- 4 Qualitative safety evaluation is permitted for non-complex equipment; a quantitative probabilistic analysis (and the DO-178C machinery that feeds it) is not required for a minor–failure advisory function¹.

Honest caveats (must stay in the doc): (1) This argument must be agreed with the FAA ACO early — the applicant proposes the classification; the FAA concurs. If the FAA judges the failure condition above minor (e.g. because of over-reliance / automation-dependency human factors, or because the audio–panel interface is judged to compromise an aircraft communication system), the program moves to §2 of the NORSEE policy (xx.1309, ARP4754A/ARP4761) and software assurance re-enters. (2) “No DO-178C” is earned by architecture and procedural mitigations, not by labeling — the revert-to-unopened behavior, the validation gate, the dark-cockpit annunciation, the receive-only isolated interface, and a mandatory limitation that the unit may not be used as a substitute for the required checklist are the price of that classification. (3) The audio–panel tap raises the installation bar. What was arguably a minor alteration (a self-contained box drawing only power) now wires into an aircraft communication system, pushing the path toward an STC or careful field-approval scrutiny of the interface on most airframes (C.3a, C.5) — do not assume a logbook-entry minor alteration any more. (4) VOX adds a human-factors failure mode. Hands-free activation can false-trigger on ambient cockpit speech, ATC audio, or crew conversation, potentially reading an unrequested checklist; it is mitigated by VAD sensitivity tuning, a bounded trigger grammar, the retained PTT override, and the standing limitation that the required checklist remains the authority (C.3a, D.3.2).

C.3a Audio–panel interface impact (the honest part)

Tapping the aircraft audio panel is the single biggest certification change in this revision. It must be argued explicitly, because it directly weakens the independence leg of C.3 and changes the installation classification. The goal is to make the interface so narrow and so demonstrably one-way that the residual failure stays minor.

What the interface is — and is not:

Property	Design commitment
Directionality	Receive-only. The device has no path to transmit, key a radio, mute, or inject audio into the panel. There is no DAC, line-driver, or PTT line going to the panel.
Isolation (analog path)	A 600 Ω aviation audio ground-loop isolation transformer (e.g. Allen Avionics AGL series) provides galvanic isolation between the panel and the device; high-impedance, line-level tap through a series resistor so the device is a negligible load.
Isolation (digital path)	A buffered, receive-only I2S input from a codec ADC fed by the same isolated / high-impedance tap; no clock or data driven back toward any aircraft bus.
Fault containment	A short, open, or power loss inside the device cannot load down, ground, or back-feed the panel — the isolation barrier and high-impedance tap see to that.
No operational credit	The panel feed is <i>listened to</i> for recognition only; it is never relied on for any aircraft function.

Why this still supports a minor classification:

- The audio panel and intercom continue to function identically whether the device is present, powered, or failed — the tap is parallel and high-impedance.
- The failure modes the tap could plausibly add (loading, ground loop, injected noise) are removed by isolation and the receive-only topology, and are exactly what DO-160G conducted/induced-susceptibility and the audio-system installation tests are meant to verify (C.4).
- This is analogous to other listen-only cockpit aids (cockpit voice recorders, audio-logging headsets) that tap audio without compromising the source.

Why it nonetheless raises the bar (do not gloss over this):

- The interface now touches an aircraft communication system, so the installation will in most cases be evaluated as a change that affects that system — pushing the path toward STC / careful field approval rather than a simple logbook minor alteration (C.5).
- The ACO may require substantiation that the tap does not degrade comm audio (intercom level, sidetone, hot-mic/VOX behavior of the *panel's own* circuits) under all conditions, including device failure.
- Audio-panel wiring practice matters: many panels ground audio jacks only at the intercom to avoid ground loops, so the tap point and shield grounding must be coordinated with the specific panel's installation manual.

VOX (voice-activation) certification note. Replacing push-to-talk with VOX as the primary trigger introduces a false-activation human-factors mode (reading an unrequested checklist on stray speech / ATC audio). Mitigations carried into the design: bounded wake-word + trigger grammar (not open-vocabulary), tunable VAD sensitivity and hangover (Config Card, B.2), the retained PTT override, and the standing limitation that the required checklist remains the authority. The ACO will want this mode addressed in the safety / human-factors evaluation¹.

C.4 DO-160G environmental qualification plan

Categories a cockpit/avionics-bay unit would target. For the prototype these are design targets, not completed tests².

DO-160G section	Target category	Rationale for this unit
§4 Temperature & Altitude	Cat A2 (controlled/pressurized)	-15 °C to +55 °C operating; pressurized cabin
§5 Temperature Variation	Cat B	Cockpit rate of change
§6 Humidity	Cat A	Conformal coat recommended
§7 Operational/Crash Shock	Operational + crash-safety	Boards retained; nothing becomes a projectile
§8 Vibration	Cat S (fixed-wing)	Locking hardware; conformal coat; no press-fit-only parts
§15 Magnetic Effect	Class Z	Small device; classify by measured deflection
§16 Power Input	per installation (e.g. 28 VDC)	Only if a 28 V variant is built; TVS + fuse
§17 Voltage Spike	Cat A	Input transient protection
§18 AF Conducted Susceptibility	Cat —	As applicable to the supply; also relevant to the audio-panel input — verify recognition is not corrupted by AF conducted noise
§19 Induced Signal Susceptibility	Cat ZC	Cockpit; audio-input cabling routed/shielded per the panel's installation practice
§20 RF Susceptibility	Cat —	Aluminum case + grounded shield
§21 RF Emission	Cat M (or better)	Wi-Fi/BT disabled in firmware materially helps emissions
§22 Lightning Induced Transient	as installed	Behind-panel mounting reduces exposure
§25 ESD	per §25	Bond exposed metal; recessed connectors
§26 Flammability	UL94 V-0 materials	Required if not aluminum

The deliberate choice to disable Wi-Fi and Bluetooth in firmware is both a security decision and an emissions-qualification advantage (§21).

Audio-interface-specific evidence (beyond the table): because the unit now taps the audio panel, qualification should additionally demonstrate that — across all DO-160G conditions and including a failed/unpowered device — the tap does not degrade audio-panel performance (intercom level, sidetone, the panel's own VOX/hot-mic behavior). The isolation transformer and high-impedance receive-only topology (C.3a) are the design basis for that demonstration.

C.5 Approval & installation path (per airframe)

NORSEE approval is design + production approval — not installation approval. The full chain for a Part 23/27/29 airframe:

- 1 Pre-application — engage the responsible FAA ACO early; agree on the NORSEE classification (minor change / minor failure) and the MDR (C.2).
- 2 Design + test — complete DO-160G qualification, the safety evaluation, human-factors assessment, and the configuration-control / card-provenance package.
- 3 NORSEE Letter of Approval (LOA) under § 21.8(d), with the certifying statement of compliance (template in C.6) and a quality system per the policy.
- 4 Installation — NORSEE *eligibility* still requires an installation approval on each aircraft: if a minor alteration, a logbook entry / FAA Form 337 as applicable; if a major change to type design (panel structure, electrical integration), an STC or field-approval path.
- 5 Part 25 aircraft — skip NORSEE; pursue STC for the installation (C.1.1).

The audio-panel interface raises the install classification. Because the device now wires into an aircraft communication system (the audio panel), step 4 should be approached assuming the interface makes the alteration more than minor on most airframes — i.e. plan for an STC or a field approval that specifically substantiates the audio tap (receive-only, isolated, no degradation of comm audio per C.3a/C.4), not a bare logbook entry. The earlier power-only/independent design could credibly claim a minor alteration; this one generally cannot.

C.5.1 Where TSO and PMA fit (and don't, here)

- TSO authorization is a *minimum-performance* design+production approval against a specific TSO. There is no TSO that defines a “voice advisory checklist reader,” so a TSOA is not the natural path; NORSEE (applicant-proposed MDR) is. A TSOA would also still require separate installation approval³.
- PMA is for *modification/replacement* articles for a type-certificated product and is typically tied to an STC or identity. This unit is added equipment, not a replacement part, so PMA is not the primary path — though a production unit may end up holding PMA in conjunction with an STC for specific airframes (14 CFR § 21.303).

C.6 Certifying statement of compliance (template)

Per the NORSEE policy, the LOA application carries a statement in this form (placeholders to be completed at application). *Reproduced as a template only — not a current certification.*

“I, (authorized representative), certify that (company) has complied with all applicable requirements as identified in RTCA/DO-160G and the other Minimum Design Requirements accepted for this article, and policy statement PS-AIR-21.8-1602, and that the article is produced under the required quality system.”

Two limitation statements the FAA expects on advisory NORSEE, to be placarded / in the manual:

“No operational credit may be taken for installation of this system.” · “This system is not a required system and may not be used as a substitution for the certificated aircraft checklist.”

C.7 Compliance summary matrix

Requirement	Means of compliance	Status (demo)
Non-required, safety-enhancing	NORSEE PS-AIR-21.8-1602, § 21.8(d)	Argued; not applied
Minor change / minor failure	Qualitative safety evaluation (C.3)	Drafted; ACO concurrence pending
Environmental	RTCA/DO-160G (C.4) per AC 21-16G	Targets defined; not tested
Software assurance	DO-178C not sought — advisory/minor (C.3)	Architecture supports it; ACO concurrence pending
Complex hardware	DO-254 not invoked (simple COTS) — AC 20-152A	N/A by design
Human factors / color	AC 25-11B conventions, dark-cockpit	Implemented in design
Audio-panel interface	Receive-only + galvanic isolation (C.3a, C.4 §18/§19)	Architecture defined; substantiation/test pending
VOX false-activation	Bounded grammar + tunable VAD + retained PTT override (C.3a/D.3.2)	Mitigations defined; ACO human-factors concurrence pending
Installation	STC / substantiated field approval (audio tap is more than minor); minor-alteration unlikely	Per-aircraft; none performed
Part 25 airframes	STC (NORSEE excluded)	Flagged

¹ FAA Policy Statement PS-AIR-21.8-1602, “Approval of Non-Required Safety Enhancing Equipment (NORSEE)” — [gajsc.org](https://www.faa.gov/gajsc)

² RTCA/DO-160G; FAA AC 21-16G acceptance of DO-160 D-G — [faa.gov AC 21-16G](https://www.faa.gov/aircraft/air_cert)

³ FAA Technical Standard Orders (TSO) program — [faa.gov/aircraft/air_cert](https://www.faa.gov/aircraft/air_cert)

PART D

Hardware Reference

The generic hardware platform shared by every aircraft card: MCU, pin map, per-device wiring, annunciator, lamp driver, power budget, processor selection, firmware notes, and the DIY bill of materials. Pin numbers match firmware/main/board_pins.h.

Pin numbers match firmware/main/board_pins.h. The wiring diagram is in §D.10. Logic level is 3.3 V (the ESP32-S3 is not 5 V tolerant on GPIO).

D.1 System overview

Item	Value
MCU	ESP32-S3 (dual-core LX7 @ 240 MHz) — PSRAM required by ESP-SR
Recommended module	ESP32-S3-WROOM-1 N16R8 (16 MB flash, 8 MB octal PSRAM)
Speech stack	Espressif ESP-SR: AFE (NS/VAD → VOX) → WakeNet “Hi ESP” → MultiNet English
Crew audio input	From the aircraft audio panel on I2S_NUM_0, selectable per install: analog (isolated line tap → I2S codec ADC) or digital (I2S codec ADC fed from a buffered tap). Onboard MEMS mic = bench-test only
Activation	VOX (AFE VAD) primary, hands-free; PTT retained as manual override
Audio output	I2S Class-D amplifier on I2S_NUM_1 → 4–8 Ω speaker (own speaker; not fed to the panel)
Config storage	two microSD slots — slot 1 Config Card, slot 2 Data Card (Part B), FAT32
Annunciation	Applied Avionics split-legend switch (dark-cockpit, AC 25–11B)

Two build paths: Integrated (ESP32-S3-Korvo-2 dev board — ES8311/ES7210 codec, NS4150 amp, microSD slot; line-in repurposed for the audio-panel feed) or DIY (ESP32-S3 DevKitC-1 N16R8 + audio-panel input stage [isolation transformer + I2S codec ADC] + MAX98357A amp + two microSD breakouts + the annunciator switch). The earlier INMP441 MEMS mic remains available only as a bench-test input.

D.2 Master pin map

Function	Macro	GPIO	Dir	Notes
PTT override	PTT_GPIO	0	in (PU)	BOOT button; active-low; force-listen override of VOX
SELECT switch	SELECT_GPIO	10	in (PU)	IN = GPIO→GND (active-low)
Legend OFF (white)	LEGEND_OFF_GPIO	21	out	top legend half (via driver)
Legend FAULT (amber)	LEGEND_FAULT_GPIO	14	out	bottom legend half (via driver)
Status LED	STATUS_LED_GPIO	48	out	on-board RGB on most S3 devkits
Audio-in bit clock	AIN_BCLK_GPIO	4	out	I2SO BCLK → codec/mic SCK

Function	Macro	GPIO	Dir	Notes
Audio-in word select	AIN_LRCLK_GPIO	5	out	I2SO WS → codec/mic WS
Audio-in data	AIN_DIN_GPIO	6	in	codec ADC / mic SD → ESP DIN
Audio-in master clock	AIN_MCLK_GPIO	3	out	MCLK to the codec (ES8388/ES7210 need it; INMP441/PCM1808 do not)
Codec I2C SDA	CODEC_SDA_GPIO	1	i/o	ES-series codec control bus (digital path)
Codec I2C SCL	CODEC_SCL_GPIO	2	out	ES-series codec control bus (digital path)
SD clock	SD_CLK_GPIO	7	out	SDMMC CLK (shared by both card slots)
SD command	SD_CMD_GPIO	9	i/o	SDMMC CMD (needs pull-up)
SD data 0	SD_D0_GPIO	8	i/o	SDMMC DAT0 (needs pull-up)
Config-card detect	SD_CFG_CD_GPIO	47	in (PU)	slot-1 card-detect (Config Card)
Data-card detect	SD_DAT_CD_GPIO	38	in (PU)	slot-2 card-detect (Data Card)
Speaker bit clock	SPK_BCLK_GPIO	15	out	I2S1 BCLK → amp BCLK
Speaker word select	SPK_LRCLK_GPIO	16	out	I2S1 WS → amp LRC
Speaker data out	SPK_DOUT_GPIO	17	out	I2S1 DOUT → amp DIN

Polarity macros: PTT_ACTIVE_LOW=1, SELECT_ACTIVE_LOW=1, LEGEND_OFF_ACTIVE_HIGH=1, LEGEND_FAULT_ACTIVE_HIGH=1, STATUS_LED_ACTIVE_HIGH=1, LAMP_TEST_MS=2000. Set any unused output to -1 to disable it cleanly.

Two-slot SD note: the demo shares one SDMMC 1-bit bus (CLK/CMD/DAT0) across both card slots, distinguished by per-slot card-detect lines and by mounting each at its own path (/sdcard-config, /sdcard-data). A production unit may instead give each slot its own SPI or SDMMC bus to remove any contention; the firmware reads Config first, then Data (B.4).

Reserved / avoid pins (N16R8): GPIO33–37 (octal PSRAM/flash bus — do not use), GPIO19/20 (USB D-/D+), GPIO0/45/46 (strapping — must boot in the right state), GPIO26–32 (SPI flash on some modules). GPIO0 here is only the BOOT/PTT button, so it is safe. GPIO1/2/3 are used here for the codec I2C + MCLK; on the Korvo-2 use that board's BSP assignments instead.

D.3 Per-device wiring

D.3.1 Audio-panel input stage (the crew-audio source, I2S_NUM_0). Crew speech comes from the aircraft audio panel, not an onboard mic. The source is chosen by the Config Card (audio_source); both paths terminate as an I2S input to the ESP32-S3 and feed the ESP-SR AFE.

- Analog path (analog). Tap a headphone/intercom/line output from the panel (aviation audio is typically ~150–600 Ω , ~1–5 V RMS). Feed it through a 600 Ω audio ground-loop isolation transformer (e.g. Allen Avionics AGL series) for galvanic isolation, then through a ~220–470 Ω series resistor + simple RC anti-alias into the line-in of an I2S codec ADC (ES8388/ES7210 with MCLK on GPIO3, I2C control on GPIO1/2; or PCM1808/CS5343 which self-clock). The tap is high-impedance and parallel so the panel sees a negligible load; the device cannot back-feed the panel. Scale the divider so panel line level maps to the codec's full-scale input without clipping.
- Digital path (digital). Where the codec sits closer to the source, take a buffered, receive-only I2S/line feed into the same codec ADC. There is no I2S output toward the panel — BCLK/WS/MCLK are generated by the ESP32 for the ADC only, and no data line is driven back toward any aircraft bus.
- Codec wiring: SCK→GPIO4, WS→GPIO5, ADC_DATA→GPIO6, MCLK→GPIO3, I2C SDA/SCL→GPIO1/2; supply per the codec (1.8–3.3 V analog/digital rails); decouple each rail 0.1 μ F.
- Bench-test option only: an INMP441 MEMS mic may be wired in place of the codec for desk testing (VDD→3V3, SCK→GPIO4, WS→GPIO5, SD→GPIO6, L/R→GND, 100 k Ω pulldown on SD, never clock with VDD off). This is not the installed audio source and must not be used in an aircraft (it does not hear the panel and breaks the receive-from-panel model).

D.3.2 VOX & PTT override. Recognition is gated by the AFE VAD (VOX): the device listens whenever speech is detected on the panel feed, tuned by the Config Card (vox.vad_sensitivity, vox.hangover_ms). The PTT button (GPIO0, active-low) is a force-listen override — holding it opens recognition regardless of VAD, and with vox.mode = ptt_only it becomes the sole trigger (VOX disabled). PTT is debounced in software. There is no PTT/keying line toward the aircraft — this button only tells the device's own recognizer to listen.

D.3.3 MAX98357A Class-D amp → ESP32-S3 (I2S_NUM_1). Supply 2.5–5.5 V; ~2.4 mA quiescent; peak ~650 mA at 5 V/4 Ω ; no MCLK. VIN→5 V (full output), GND→GND, BCLK→GPIO15, LRC→GPIO16, DIN→GPIO17, GAIN NC = 9 dB, SD/mode float = mono. OUT+/OUT- are bridge-tied — never to GND. The device drives its own speaker; its output is never routed back into the audio panel.

D.3.4 Two microSD card slots → SDMMC 1-bit. 3.3 V cards. Shared bus CLK→GPIO7, CMD→GPIO9 (10 k Ω →3V3), DAT0→GPIO8 (10 k Ω →3V3), VDD→3V3, VSS→GND. Slot 1 = Config Card (detect on GPIO47, mount /sdcard-config), slot 2 = Data Card (detect on GPIO38, mount /sdcard-data). FAT32; layouts per Part B. (Production option: separate bus per slot.)

D.3.5 Discrete inputs. SELECT (GPIO10): LOW = selected IN; pull-up HIGH = OUT. PTT (GPIO0): LOW = pressed (override). Each is a simple SPST to GND; debounce in software.

D.4 Annunciator switch (split-legend, dark-cockpit)

State	TOP — white VOICE CHKLST OFF	BOTTOM — amber VOICE CHKLST FAULT
Selected IN, healthy	dark	dark ← true dark cockpit
Selected IN, fault	dark	amber ON

State	TOP — white VOICE CHKLST OFF	BOTTOM — amber VOICE CHKLST FAULT
Selected OUT	white ON	dark (fault inhibited)

Power-up lamp test: both halves on for ~2 s, then dark, so a dead LED cannot masquerade as healthy. OUT inhibits fault per AC 25-11B — a deselected system needs no crew action, so only the white OFF status shows; recognition and fault monitoring are suspended while OUT.

D.5 Lamp-driver circuit (one per legend half)

An ESP32 GPIO (~20 mA default, 40 mA max; 1.5 A total chip limit) cannot drive a 28 V — or 5 V at lamp current — legend directly. Use a low-side switch per half:

```

+V_lamp (5 V or 28 V, separate rail)
|
[ legend lamp ]      <- VIVISUN legend (white or amber)
|
+----- Drain
GPIO21 --[1k]--|G  N-ch MOSFET (logic-level, e.g. 2N7002 / A03400)
(or G14)         |  or NPN (e.g. 2N2222 with base resistor)
[10k]            <- gate/base pulldown -> GND (defined OFF)
|
GND (Source) ----- common ground with ESP32

```

One driver for GPIO21 (OFF/white), one for GPIO14 (FAULT/amber). The 10 kΩ pulldown guarantees the lamp is OFF during boot/reset before the GPIO is configured (dark-cockpit integrity). For a 28 V incandescent legend, use a FET with $V_{ds} \geq 40$ V and current $\geq$ inrush. Applied Avionics VIVISUN/Korry legends come in 28 VDC, 5 VDC, 28 VAC, 5 VAC, and 115 V lamp variants. For a quick bench mock-up, substitute two 3.3 V LEDs (white + amber) with series resistors driven straight from GPIO21/GPIO14 within the ~20 mA limit.

D.6 Power budget

Rail	Loads	Typical	Peak
3.3 V	ESP32-S3 (Wi-Fi off) + audio codec + two microSD slots	~90–170 mA	~280 mA (SD init / SR burst)
5 V	MAX98357A output	a few mA idle	~650 mA (5 V/4 Ω , loud)
Lamp rail	up to 2 legend halves	0 (dark)	per lamp spec (e.g. 28 V incand.)

The audio-input codec (a few mA—~20 mA) and the second microSD slot add a little to the 3.3 V rail; the isolation transformer is passive. Power from USB 5 V $\geq$ 1 A. Keep a 28 V legend supply separate from logic 5 V (grounds common only). Bulk decoupling $\geq$ 100 μ F near the amp VIN plus 0.1 μ F per device. The audio-panel tap draws no power from the aircraft and is isolated from the device's own grounds through the transformer (analog path).

D.7 Processor selection

This is a narrow, bounded workload — a small fixed vocabulary (checklist triggers + universal advance words, well under the ~200-command MultiNet cap), offline, and deterministic. That is command recognition on a fixed grammar, not open-ended transcription, which keeps an MCU-class part firmly in scope and makes a Linux SBC overkill. It is also exactly the property that supports the minor-failure / no-DO-178C argument (C.3): a single-chip, fixed-grammar device has a small, well-understood failure surface.

Recommendation: keep the ESP32-S3 as the baseline; consider the ESP32-P4 only for headroom. ESP-SR (v2.1+) supports S3 and P4 for English MultiNet; the classic ESP32 is no longer supported by the current speech algorithms and should be avoided.

Option	Summary	Verdict
ESP32-S3 (baseline)	LX7 dual-core @240 MHz + AI vector ext, 512 KB SRAM, Wi-Fi+BLE, ~\$8–15	Best AI/voice part in the line, most mature tooling; firmware/diagram/BOM built on it
ESP32-P4 (upgrade)	RISC-V to 400 MHz + AI ext, 768 KB SRAM, ~2.5 $\times$ compute, no Wi-Fi/BT	Worth it only for more compute or a future display; no-radio is arguably a safety plus
Syntiant NDP120	Always-on ultra-low-power NDP	Overkill; we have panel power, need full grammar + playback + SD
Picovoice Porcupine+Rhino	Offline wake+intent on Cortex-M4	Clean fit but needs a per-deployment license key — undesirable for self-contained safety gear
Fluent.ai / NXP i.MX RT600	Commercial intent / heavy audio DSP	More licensing/board complexity than a fixed grammar needs
Raspberry Pi (whisper/Vosk)	Full Linux STT	Non-deterministic boot, higher power/cost — less robust for a fixed-grammar safety device

D.8 Firmware build notes

Topic	Value
Framework	ESP-IDF ≥ 5.2
Components	esp-sr, esp_spiffs, driver, json (cJSON), fatfs, sdmmc, esp_driver_sdmmc, esp_codec_dev (codec init for the digital/analog path)
Speech models	WakeNet WN9_HIESP; MultiNet English mn6_en/mn7_en (S3 only)
Partitions	factory app 3 MB + model 5 MB + storage 2 MB → needs 16 MB flash (N16R8)
Audio input	ESP-SR AFE fed from I2SO (codec ADC); VAD→VOX gating, PTT override; codec init from Config Card audio.codec
Two-card load	Config Card first (/sdcard-config/config.json) → init audio source + VOX → then matching Data Card folder (/sdcard-data/<aircraft>/)
Grammar rules	lowercase + single spaces; spell numbers (“v one”); ~200-cmd cap
Fault behavior	any card/config fault → ST_FAULT, amber legend, no checklist shown (B.4)

D.9 Bill of materials (DIY build)

Qty	Part	Spec / example
1	ESP32-S3 DevKit	DevKitC-1 N16R8 (PSRAM)
1	Audio-panel input codec	I2S codec ADC with line-in: ES8388 / ES7210 (need MCLK+I2C) or PCM1808 / CS5343 (self-clocking)
1	Audio isolation transformer	600 Ω:600 Ω aviation audio ground-loop isolator (Allen Avionics AGL series)
1	Input network	~220–470 Ω series resistor + RC anti-alias for the analog tap
1	I2S amp	MAX98357A breakout
1	Speaker	4–8 Ω, ≥ 2 W
2	microSD card + breakout	FAT32 — Config Card (slot 1) + Data Card (slot 2)
1	Annunciator switch	Applied Avionics VIVISUN/Korry split-legend (or 2 LEDs for bench)
2	Lamp driver	logic-level N-MOSFET (2N7002/AO3400) or NPN (2N2222)
4	Resistors	1 kΩ ×2 (gate), 10 kΩ ×2 (pulldown)
2–4	Pull-ups	10 kΩ on SD CMD/DAT0; codec I2C pull-ups if needed
—	Caps	0.1 μF per device/rail, 100 μF bulk near amp
1	PTT button	momentary SPST (or use BOOT) — VOX override
(opt.)	INMP441 MEMS mic	bench-test input only, not the installed source

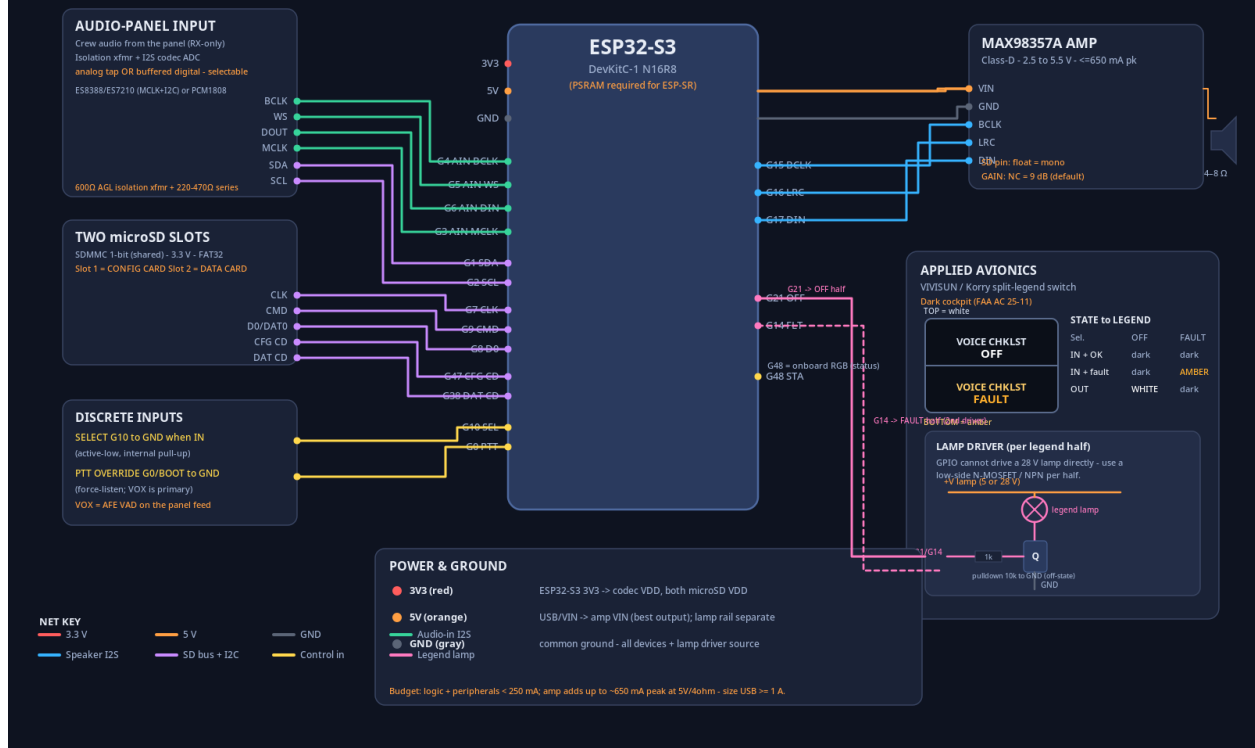
Integrated alternative: ESP32-S3-Korvo-2 (~\$45–55) provides codec/amp/SD on-board; use its BSP pin map and codec (ES8311/ES7210) init, and repurpose its line-in for the audio-panel feed. A production unit adds the second card slot.

D.10 System wiring diagram

CJ2 Voice Emergency Checklist — DIY Wiring Diagram

ESP32-S3 + audio-panel input (isolated I2S codec) + MAX98357A amp + two microSD (Config + Data) + Applied Avionics split-legend switch

Pin numbers match `firmware/main/board_pins.h` • DEMO / TRAINING ONLY — NOT FOR FLIGHT



Complete system wiring — ESP32-S3, audio input, amp, two microSD cards (Config + Data), and split-legend annunciator with per-half lamp drivers.

PART E

Enclosure Specification

For the fabricating engineer. The housing is generic — sized for the electronics and the panel interface, not for a specific airframe. The only airframe-specific item is the mounting variant (DZUS slot vs. round instrument hole) and the card inside it.

E.1 Two-piece architecture

Piece	Contents	Where	Why
A. Panel bezel	Split-legend annunciator switch, speaker + grille, PTT override button	Front panel / pedestal, on the DZUS rail	Crew must see/reach it; dark-cockpit annunciator in the normal scan
B. Remote box	ESP32-S3, audio-input stage (isolation transformer + codec), amp, two microSD card slots, lamp-driver, power conditioning	Avionics bay / behind-panel, blind	Keeps heat, the card slots, and wiring out of the panel; close to the audio-panel tap point

A single all-in-one box is acceptable for a pure bench demo, but the two-piece split mirrors real remote-mount avionics and keeps the audio-input stage near the panel tap.

E.2 Piece A — panel bezel

- DZUS rail mount: fastener pitch 3/8 in (9.525 mm); clearance hole 0.255 in (6.48 mm); bezel height a whole multiple of 3/8 in (target 3-unit = 28.575 mm, or 4-unit = 38.1 mm if the grille needs room); standard pedestal width $\approx$ 146 mm aluminum (144.45 mm face); backplate 1/16 in (1.6 mm) 6061-T6; first/last fastener 14.29 mm from each end.
- Round-hole variant: fits a standard 3-1/8 in (79.4 mm) instrument cutout with four 6-32 screws on the standard bolt circle, to replace a blanking plate where no DZUS slot exists.
- Face layout (top→bottom): split-legend annunciator switch (cut per the *specific* switch datasheet — typical VIVISUN bezel $\approx$ 15×15 mm to 19×19 mm; top half VOICE CHKLST OFF white, bottom VOICE CHKLST FAULT amber, upright when installed); speaker grille ($\geq$ 40 % open over the cone, offset from the switch); optional guarded/recessed PTT.
- Material/finish: 6061-T6 aluminum 2.0–3.0 mm (or ABS/PC for a non-structural demo); matte black, low-gloss ($\leq$ 10 gloss units) to suppress glare; legend by the switch's internal engraving (preferred) or laser-etch + white/amber paint-fill; edges chamfered 0.5 mm.

E.3 Piece B — remote processor box

- Envelope: sized around the ESP32-S3 DevKitC-1 ($\approx$ 70×26 mm) plus amp, the audio-input stage (codec + isolation transformer), two microSD breakouts, and the 2-channel lamp-driver; practical outer $\approx$ 120 × 85 × 45 mm (slightly larger than before to fit the transformer + second slot). Confirm against the actual stacked board set.

- Mounting: internal standoffs / M2.5 brass inserts — boards screwed down, not floating (vibration). Keep the audio-input stage and its shielded cabling away from the amp and the switching DC-DC; the isolation transformer mounts solidly (it is a magnetic part).
- Card slots: two externally-swappable microSD carriers, clearly and distinctly labeled CONFIG CARD – FAT32 (slot 1) and DATA CARD – FAT32 (slot 2), keyed or spaced so they cannot be confused/swapped; both swappable without opening the box. The Config Card carrier should accept a write-protect-locked card.
- Access & connectors: covered/recessed USB-C service port (bench use only); one keyed, positive-latching main connector (small MIL-circular or D-sub) carrying SELECT, both legend drives, PTT, speaker +/-, power/ground (pinout from board_pins.h); plus a separate, shielded, clearly-labeled AUDIO IN (ISOLATED, RX ONLY) connector for the audio-panel tap — kept on its own keyed connector so it can never be mis-mated to power or the speaker, with the isolation transformer inside the box on the panel side of the codec. Accept USB 5 V $\geq$ 1 A; optional internal 28 V $\rightarrow$ 5 V DC-DC ($\geq$ 2 A) with TVS + fuse if a 28 V bus mock-up is wanted (mark as demo regulator, not DO-160 qualified).
- Material/EMI: aluminum preferred (doubles as EMI shield + heatsink); if plastic, add a grounded conductive shield liner/coating; single-point chassis ground stud bonded to the connector shell and ESP32 ground. Route the audio-in shield per the panel's grounding practice (often grounded only at the intercom — see C.3a).

E.4 Audio, thermal, environmental, labeling

- Audio: 4–8 Ω , ≥ 2 W speaker, sealed-back or small rear volume (5–15 cm³); grille ≥ 40 % open with acoustic mesh; gasket the speaker to prevent buzz.
- Thermal: ESP32-S3 + ESP-SR is low-power (a few hundred mW) — no fan. Passive convection (vent slots low/high) or conduction (thermal pad to the aluminum wall). If sealed, verify internal rise < 20 °C above 55 °C ambient. DC-DC (if fitted) on its own thermal path.
- Environmental: design toward the DO-160G categories in C.4 (Cat A2 temp, Cat S vibration, etc.) — design guidance for the prototype, formal test for a productized unit.
- Labeling: placard DEMO / TRAINING ONLY — NOT FOR FLIGHT; box exterior carries unit name, serial/asset field, the two card-slot labels (CONFIG CARD / DATA CARD, FAT32), the AUDIO IN — ISOLATED, RX ONLY connector marking, and the USB “bench use only” note; annunciator legends VOICE CHKLST OFF (white) / VOICE CHKLST FAULT (amber); amber = caution, white = status per AC 25-11B.

E.5 Deliverables & open items for the engineer

Deliverables: STEP + native 3D CAD of both pieces (boards + switch modeled in place); fully-dimensioned 2D drawings (DZUS pattern, switch cutout from the chosen datasheet, grille, connector cutouts; GD&T on the switch cutout and DZUS holes); connector pinout mapped to board_pins.h; an FDM/SLA printable prototype for fit-check; a mechanical BOM; tolerances (switch cutout ± 0.1 mm, DZUS holes ± 0.1 mm on the 9.525 mm pitch, general ± 0.25 mm).

Open items (confirm before CAD): the exact Applied Avionics switch part number — the single most critical dimension; nothing finalizes until it is fixed. Also: DZUS slot vs. 3-1/8 in round hole in the target panel; the audio-panel tap point, level, and grounding for the target installation (sets the isolation-transformer + divider design); the codec part for the digital path (sets the I2C/MCLK init); whether a 28 V input is wanted; and the speaker model (sets grille open area + rear-volume cavity).

Reference dimensions: DZUS pitch 9.525 mm · DZUS hole 6.48 mm · backplate 1.6 mm · first fastener offset 14.29 mm · pedestal panel width ≈ 146 mm · round instrument hole 79.4 mm · remote box $\approx 120 \times 85 \times 45$ mm · two microSD slots · isolated RX-only audio-in connector · speaker 4–8 Ω ≥ 2 W.

PART F

References

All cited regulatory, component, and mechanical sources, with clickable links.

Certification & regulatory

#	Reference	Link
1	FAA Policy Statement PS-AIR-21.8-1602 — “Approval of Non-Required Safety Enhancing Equipment (NORSEE)”	open
2	FAA Technical Standard Orders (TSO) program overview	open
3	FAA AC 21-16G — acceptance of RTCA/DO-160 versions D–G	open
4	RTCA DO-160G — Environmental Conditions and Test Procedures for Airborne Equipment	open
5	FAA AC 20-152A — Development Assurance for Airborne Electronic Hardware (simple-hardware relief)	open
6	FAA AC 20-168 / RTCA DO-313 — installation of non-essential, non-required equipment	open
7	TSO / TC / STC / PMA primer (Afuzion)	open
8	FAA AC 25-11B — electronic flight displays; annunciation color / dark-cockpit conventions	open

Hardware & components

#	Reference	Link
1	Allen Avionics AGL audio ground-loop isolation transformers (analog audio-panel tap)	open
2	Everest-Semi ES8388 audio codec (line-in I2S ADC, MCLK + I2C)	open
3	ES7210 multichannel audio ADC (Espressif-supported codec)	open
4	TI PCM1808 stereo audio ADC (self-clocking I2S input)	open
5	Espressif ESP-SR AFE / VAD (voice-activity detection for VOX)	open
6	INMP441 microphone datasheet (bench-test option only)	open
7	MAX98357A amplifier datasheet (Analog Devices)	open
8	MAX98357A breakout guide (Adafruit)	open
9	ESP32-S3 GPIO drive current discussion	open
10	microSD operating current	open
11	Applied Avionics VIVISUN lighted pushbutton switches	open
12	Espressif ESP-SR speech framework	open
13	Espressif ESP32-P4 product page	open
14	ESP32-P4 vs ESP32-S3 performance	open

Mechanical / panel

#	Reference	Link
1	DZUS panel-building dimensions guide (MyCockpit)	open
2	Standard 3-1/8 in instrument cutout (Aircraft Spruce)	open

DEMO / TRAINING ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS. Repository: github.com/flas-tech/cj2-voice-emergency-checklist (MIT License). The certification sections describe a path, not held approvals; no NORSEE LOA, STC, PMA, or TSOA exists for this article.